Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/18/2024
PI Name:	McGee-Lawrence, Meghan Ph.D.		
Project Title:	Osteocyte Plasma Membrane Disruptions in Skeletal Adaptation to Loading and Unloading		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	12/01/2020	**End Date:**	11/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per NSSC information (Ed., 11/1/24). NOTE: End date changed to 11/30/2024 per PI and NSSC information (Ed., 10/26/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hamrick, Mark Ph.D. (Augusta University Research Institute, Inc.) Johnson, Maribeth M.S. (Augusta University Research Institute, Inc.)		
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Performance Goal No.:			
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Task Description:	The skeleton's ability to adapt to mechanical loading is crucial for bone health, as exercise promotes hypertrophy but disuse (such as from spaceflight) leads to bone loss. We were the first to report that small, transient plasma membrane disruptions (PMD) develop with in vitro and in vivo mechanical loading in bone osteocytes. These disruptions initiate skeletal mechanotransduction, suggesting PMD are stimuli recognized by osteocytes to regulate bone adaptation to its loading environment. Importantly, we consistently observe that ~20% of long bone osteocytes develop PMD with routine cage activity in mice, suggesting that formation of osteocyte PMD may be essential to bone's sensation of and response to normal gravitational loads. Accordingly, our central hypothesis is that osteocyte PMD formation is impaired during skeletal disuse, leading to bone loss. Our goals are to test the effects of disuse on osteocyte PMD formation, to determine whether osteocytes become sensitized to PMD formation with impaired PMD repair or survival during reloading, and to determine whether modulating osteocyte PMD formation and/or repair affect these processes. Our strategy is to test these concepts in an in vivo murine model of hindlimb unloading, as well as with in vitro osteocyte models of unloading (rotating wall vessel bioreactor) and reloading (fluid shear stress). Our goals align with the NASA Space Biology program as they target Research Topic 3 (Animal Biology Studies in support of Human Space Exploration)/ Sub-Topic AH1-E (Effects of fractional gravity provided by spaceflight centrifugation or ground microgravity/partial gravity analogs to gain insights into mechanisms of how animals sense, respond, and adapt to gravity shifts that are less than 1G) by discovering the contribution of osteocyte PMD formation (and hypothesized impairment during disuse) to the skeleton's adaptation to its loading environment. This project will yield a new understanding of how complex organisms adapt to the space environment, using a ground-based analog for disuse from spaceflight; we anticipate that derived data will advance strategies for skeletal maintenance and prevention of bone fractures during disuse to promote and support human space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disuse-induced bone loss, which occurs during prolonged exposure to microgravity during spaceflight and predisposes astronauts to risk of skeletal fractures, also occurs frequently on Earth in patients with spinal cord injuries, patients subjected to chronic/long-term bed rest, and in other cases of long-term decreased mobility. Furthermore, it is well understood that mechanical loading of the skeleton through physical exercise is beneficial for bone health across a wide spectrum of human patients, but there exists a substantial proportion of the population who cannot undertake regular vigorous exercise for a variety of reasons, including underlying health conditions, time constraints, or financial concerns. Therefore, understanding the fundamental mechanisms behind how bone senses and responds to changes in mechanical loading, and exploring ways to alter the skeletal response to a given level of mechanical loading (or withdrawal of loading), may lead to therapeutic interventions for improving bone health and reducing fracture risk.
Task Progress:	In the fourth year of this grant (No Cost Extension / NCE), we worked to complete the majority of the Specific Aims of the proposal. Major accomplishments include: finishing in vivo studies on PMD repair-deficient Prkd1 and PMD-susceptible Sptbn1 conditional knockout animals, and initiating the final set of disuse-related experiments aimed at conclusively establishing the role of disuse on PMD sensitization in vivo. These experiments are described below. We continued experiments inducing disuse in vitro using the Synthecon slow turning lateral vessel (rotating wall vessel) bioreactor system. Our data demonstrate that three days of culture in the bioreactor, which sufficiently up-regulates disuse-associated genes like sclerostin, sensitizes osteocytes to the formation of PMD during re-loading, as significantly more PMD are formed with application of mechanical loading to disuse-exposed cells, as compared to application of loading to static control cultures. We are now wrapping up in vivo disuse studies and mechanical loading studies in Wildtype mice, correlating PMD formation with gene expression in the wounded cells using in situ hybridization techniques (RNAscope) and immunostaining. We also published two peer-reviewed manuscripts on our in vivo genetic models developed to alter PMD responses. Using in vitro cultures, we built upon the paradigm by which PMD affect osteocyte mechanotransduction, showing that application of PMD-inducing turbulent fluid shear up-regulated expression of the mechano-sensitive gene Ptgs2/Cox-2. This gene was also increased by inhibition of the protein Prkd1, a protein critically involved in membrane repair. Our model of defective PMD repair in osteocytes, established by genetic knockout of Prkd1 in osteocytes in vivo, demonstrated impaired PMD responses and impaired adaptation to loading consistent with a critical role for PMD-mediated mechanisms in bone mechanobiology. Interestingly, treating these mice (in vivo) or immortalized or primary osteocytes (in vitro) with an FDA approved drug agent that enhances membrane stability excitingly rescued repair defects in vitro and rescued the defects in loading-induced increases in bone density in vivo, but surprisingly further exacerbated the deficiencies in response to mechanical loading. While this was an unexpected result, careful review of the literature suggested unanticipated effects on cell energetics (respiratory change inhibition in mitochondria) which may have precluded its in vivo efficacy. These findings were published in the peer-reviewed journal Bone. [Ed. Note: See Bibliography for full citations.] We also published a second manuscript on our PMD-susceptible Sptbn1 conditional knockout mouse model (Calcified Tissue, 2024), demonstrating the impact of increasing PMD susceptibility and size using in vitro and in vivo models. We continue to support the professional development of students, having supported a total of four PhD students, five medical students, three undergraduate students, and a high school student during the completion of our funded experiments. Our final PhD student defended and successfully graduated in January, 2024. All of the students involved have received authorship on either journal manuscripts or conference abstracts stemming from their contributions. Therefore, this grant has extensively supported the career development of the next generation of scientists.
Bibliography Type:	Description: (Last Updated: 11/12/2025)
Articles in Peer-reviewed Journals	Tuladhar A, Shaver JC, McGee WA, Yu K, Dorn J, Horne JL, Alhamad DW, Hagan ML, Cooley MA, Zhong R, Bollag W, Johnson M, Hamrick MW, McGee-Lawrence ME. "Prkd1 regulates the formation and repair of plasma membrane disruptions (PMD) in osteocytes." Bone. 2024 Sep;186:117147. https://doi.org/10.1016/j.bone.2024.117147 , Sep-2024
Articles in Peer-reviewed Journals	Hagan ML, Tuladhar A, Yu K, Alhamad DW, Bensreti H, Dorn J, Piedra VM, Cantu N, Stokes EG, Blumenthal D, Roberts RL, Balayan V, Bass SM, Dickerson T, Cartelle AL, Montesinos-Cartagena M, Awad ME, Castro AA, Garland T Jr, Cooley MA, Johnson M, Hamrick MW, McNeil PL, McGee-Lawrence ME. "Osteocyte Sptbn1 deficiency alters cell survival and mechanotransduction following formation of plasma membrane disruptions (PMD) from mechanical loading." Calcif Tissue Int. 2024 Sep 14:1-9. https://doi.org/10.1007/s00223-024-01285-2 , Sep-2024